EMERSON HERRERA TOLENTINO

SOFTWARE ENGINEER · MACHINE LEARNING & GENERATIVE AI

Porto Alegre, Brazil – Peru | +51 923 752 936 | +55 51 98128-9816
emersonheto@gmail.com | linkedin.com/in/emerson-herrera-tolentino

ACADEMIC STATEMENT

I am a Software Engineer with more than 8 years of professional experience, seeking to deepen my background in Machine Learning and Generative Artificial Intelligence through the Master's program in Computer Science at PPGC – UFRGS, where I am currently enrolled as a special student. My goal is to bring together my experience in scalable software architecture and advanced machine learning techniques to develop intelligent systems applied to real-world problems and to contribute to applied research, continuing the line of work started with the paper on vision-language models presented at the LXCV workshop at CVPR 2026.

RESEARCH INTERESTS

Machine Learning · Deep Learning · Generative AI and Large Language Models (LLMs) · Retrieval-Augmented Generation (RAG) · Multimodal systems and vision-language models · Model evaluation · Applied AI.

EDUCATION

Special Student — Graduate Program in Computer Science (PPGC)

UFRGS, Brazil · CMP263 Machine Learning · applying to the Master's program — *2026/2 — Ongoing*

AI Engineer Specialization Program — Diploma of Excellence (19/20)

Data Science Research Perú · 92 academic hours — *Mar 2026 — Aug 2026*

- RAG, agents, MCP, multimodal applications, LLM evaluation, guardrails and LLMOps; final project: a multi-agent system applied to enterprise processes.

Bachelor's Degree in Software Engineering

UTP — Universidad Tecnológica del Perú — *2021 — 2024*

- Degree conferred on February 6, 2025.

Master's in Microservices with Java and Spring Boot

Lite Thinking — *Jun 2024 — Sep 2025*

Software Development Technician (licensed) — 3-year technical degree

SENATI — *Mar 2016 — Jul 2019*

Computer Hardware and Networks Specialist (licensed)

SISE — *2011 — 2012*

PUBLICATIONS / RESEARCH

Co-author — "Benchmark of Multimodal Vision-Language Models for Damage Detection"

LatinX in CV (LXCV) Workshop, CVPR · Denver, USA — *Jun 2026*

- Evaluated CLIP, SigLIP, SigLIP2 and FuseLIP for disaster damage detection from images and text; fine-tuning raised SigLIP's F1 from 0.19 to 0.90. With Mariana Risco Cosavalente and Prof. Claudio Jung (UFRGS).
- Extended abstract accepted through double-blind peer review. Compared contrastive fine-tuning, two-stage few-shot adaptation and multimodal fusion on CrisisMMD and DamageDataset; combining image and text consistently outperformed either modality alone (github.com/emersonheto/damage_dataset_fewshot_vlms).

RESEARCH AND MACHINE LEARNING PROJECTS

Machine Learning Course Project (CMP263)

PPGC – UFRGS · Python, Jupyter *2026/2*

- Team project with real-world datasets; the course covers preprocessing, class imbalance, dimensionality reduction, ensemble methods and support vector machines.

Certilab Agentic RAG — Calibration Certificate Assistant

LangGraph, FastAPI, Qdrant, OpenAI *2026*

- Agentic RAG service combining structured and semantic search, with cited sources and access control; 7-node Adaptive RAG variant with two self-correction loops over 3,848 vectors (github.com/emersonheto/certilab-agentic-rag).

Supervised Classification Models

Python, Pandas, Scikit-Learn *2025*

- Breast cancer classification (Breast Cancer Wisconsin) with cleaning, normalization, EDA and KNN; smoking-status prediction from biosignals with a decision tree (entropy, depth 5).

PROFESSIONAL EXPERIENCE

Analyst Programmer II | Full Stack AI Engineer / GenAI & RAG Developer

Canvia · Peru (remote) *Oct 2025 — Present*

- Client: Central Reserve Bank of Peru (BCRP). Designed, built and improved a multi-tenant RAG platform serving 4 of the bank's applications (BCRP Portal, Research Portal, NEXO 360 and Banknotes), including the ingestion and indexing of thousands of documents and hundreds of thousands of chunks; its assistant is live on the Research Portal (investigacion.bcrp.gob.pe).
- RAG pipelines for research articles and institutional documents, with multimodal ingestion of PDFs and images, vector search on Oracle 23ai, pgvector, Pinecone and Chroma, and LLMs through the OpenAI API and LangChain.

Full Stack & AI Developer (Freelance)

Certilab S.A.C. · Remoto *Sep 2024 — Present*

- Certification system in Laravel 11 and Filament with 300+ certificates issued: admin back office, client intranet to look up their own certificates, and a public QR validation portal with encrypted IDs.
- RAG and chatbot over calibration certificates with multimodal ingestion and a query dashboard for management; agentic version with LangGraph, FastAPI and Qdrant (github.com/emersonheto/certilab-agentic-rag).

Full Stack Developer | Angular, Java, .NET & Process Automation

Way Tech (WIN Empresas / Universidad Continental projects) · Peru (remote) *Nov 2024 — Oct 2025*

- Built academic and institutional platforms for Universidad Continental (24,000+ students across Huancayo, Arequipa, Cusco and Lima) with Angular and microservices in C# (.NET) and Java.

Systems Engineer

Indra · San Isidro, Peru (remote) *Mar 2021 — Aug 2024*

- Client: Telefónica del Perú. Distributed systems and microservices for banking and telecommunications environments.
- Real-time processing of home modem telemetry across Peru by zone: Python ETL microservice integrating network status from several APIs, raising alerts and feeding customer management and monitoring dashboards.

IT Systems Analyst

Multilab · Pueblo Libre, Peru *Feb 2020 — Mar 2021*

- Transactional clinical systems with Laravel and CodeIgniter: new features, reporting, and roles and permissions.

Software Development Area Manager · Software Developer

MControl Energy Technology SAC · Lima, Peru *Nov 2017 — Jan 2020*

- Area manager (Jul 2019 — Jan 2020): led a team of 2 developers and projects for service stations of the Gasocentro and Irasa groups in Lima and Ica. Developer (2017 — 2019): C#, ASP.NET, Laravel and Vue.js on SQL Server and MySQL.

TECHNICAL SKILLS

Generative AI & Agents: RAG (agentic and adaptive), LangChain, LangGraph, OpenAI API, Prompt Engineering, embeddings, vector databases (pgvector, Pinecone, Chroma, Qdrant), multi-agent architectures, MCP (Model Context Protocol), LLM evaluation and LLMOps

Machine Learning: Scikit-Learn, Pandas, NumPy, PyTorch, vision-language models (CLIP, SigLIP), fine-tuning and few-shot

Research: experimental design and model benchmarking, evaluation metrics (F1, calibration), Jupyter, LaTeX

Languages & Frameworks: Python (FastAPI), Java (Spring Boot), C# (.NET), Node.js, PHP (Laravel, CodeIgniter), JavaScript/TypeScript

Databases: PostgreSQL, Oracle, SQL Server, MySQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Azure, Docker, Kubernetes, CI/CD (GitHub Actions), Dokploy, Linux (Red Hat, Ubuntu)

AI-assisted development: coding agents (Claude Code, Codex, Pi) with Gentle Shell, Gentle AI and Engram persistent memory, ODD workflow

CERTIFICATIONS

- **Data Science Research Perú (2026), Diplomas of Excellence:** Evaluation, Observability and LLMOps (19) · Multimodal Agents (19) · AI Agents and MCPs, Agentic RAG (18) · Prompt Engineering and RAG (18) · AI Engineering Fundamentals (18) · Natural Language Processing.
- **Tecylab:** DevOps Multicloud Specialist (2025) · Angular Specialist (2025) · Senior in NodeJS Microservices with Cloud (2025) · Senior in Spring Boot Programming on Multicloud (2024).
- Google Cloud Platform — DSRP (2024) · Scrum Foundation Professional Certification (SFPC™) — CertiProf (2024)

LANGUAGES

Spanish: native · Portuguese: B2 · English: intermediate (advanced technical)